

Flexible vs Inflexible Methods

Bias–Variance Tradeoff in ISLR Chapter 2

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Question 1: Flexible vs Inflexible Methods

For each of parts (a) through (d), indicate whether we would generally expect the performance of a **flexible statistical learning method** to be **better or worse** than an **inflexible method**. **Justify your answer.**

- (a) The sample size n is extremely large, and the number of predictors p is small.
- (b) The number of predictors p is extremely large, and the number of observations n is small.
- (c) The relationship between the predictors and the response is highly non-linear.
- (d) The variance of the error terms, i.e. $\sigma^2 = \text{Var}(\varepsilon)$, is extremely high.

Key Concepts: Flexibility, Bias, Variance, Overfitting

- **Flexible Methods:** - Can adapt to complex patterns in data. - Low bias, high variance. - Risk: Overfitting (fits noise as well as signal). - Examples: Splines, Boosting, KNN with small K .
- **Inflexible Methods:** - Strong assumptions, simple functional form. - High bias, low variance. - Stable and interpretable. - Example: Linear Regression.
- **Bias:** Error from oversimplifying the model. High bias $\rightarrow$ misses true patterns.
- **Variance:** Error from sensitivity to small changes in data. High variance $\rightarrow$ model shape changes easily.
- **Overfitting:** Model fits both signal and noise. Usually happens with flexible methods, leading to poor generalization.

Question (a)

Scenario:

- Sample size n is extremely large
- Number of predictors p is small

Preferred Method: Flexible

Justification for (a)

- Large n provides enough information to estimate complex patterns.
- The risk of overfitting decreases as noise averages out.
- Flexible models can learn the true underlying structure safely.

Conclusion: Flexible methods perform **better**.

Question (b)

Scenario:

- Number of predictors p is extremely large
- Sample size n is small

Preferred Method: Inflexible

Justification for (b)

- Flexible models have many parameters to estimate.
- With small n , they tend to memorize noise.
- Inflexible models reduce variance and avoid overfitting.

Conclusion: Flexible methods perform **worse**.

Question (c)

Scenario:

- The relationship between predictors and response is highly non-linear

Preferred Method: Flexible

Justification for (c)

- Inflexible methods assume linearity.
- This leads to high bias when the true relationship is curved.
- Flexible methods adapt to complex patterns.

Conclusion: Flexible methods perform **better**.

Question (d)

Scenario:

- Error variance σ^2 is extremely high

Preferred Method: Inflexible

Justification for (d)

- High noise causes jagged, unstable patterns.
- Flexible models try to fit random fluctuations.
- Inflexible models focus on the overall trend.

Conclusion: Flexible methods perform **worse**.

Summary Table

Scenario	Preferred Method	Key Reason
Large n , Small p	Flexible	Low overfitting risk
Small n , Large p	Inflexible	Controls variance
Non-linearity	Flexible	Reduces bias
High noise (σ^2)	Inflexible	Avoids fitting noise

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: With Applications in R* (2nd ed.). Springer.

Relevant Sections:

- Chapter 2: Statistical Learning
- Section 2.1.3: Prediction vs Interpretability
- Section 2.2.2: Bias–Variance Tradeoff
- Figure 2.7: Flexibility vs Interpretability

Thank You